

Formal verification - Tutorial 11

01/06/2026

Finite Markov chains

Exercise 1. Let $M = (S, \rightarrow)$ be a finite Markov chain and consider a target set $T \subseteq S$. For every initial state $s \in S$, let

$$x_s = \mathbb{P}(M, s \models \mathbf{F} T)$$

be the probability of reaching a state in T when started from s .

1. Can we compute the set of initial states s such that $x_s = 0$ without solving linear equations?
2. Can we compute the set of initial states s such that $x_s = 1$ without solving linear equations?

Exercise 2. Let $M = (S, \rightarrow)$ be a finite Markov chain and consider two sets $I, T \subseteq S$. For every initial state $s \in S$, let

$$x_s = \mathbb{P}(M, s \models I \mathbf{U} T)$$

be the probability of reaching a state in T while staying in I when started from s .

1. Provide a characterisation of the vector of probabilities $(x_s)_{s \in S}$.
2. Can we compute these probabilities?

Exercise 3. Let M be a Markov chain and let L be an ω -regular property of M recognised by a nondeterministic Büchi automaton A . Can we compute $\mathbb{P}(M \models L)$ by looking at the synchronised product $M \times A$?

Exercise 4. Let M be a Markov chain and let L be an ω -regular property of M recognised by an unambiguous Büchi automaton A . Show that $\mathbb{P}(M \models L)$ can be computed in *PTIME*.

Random walks

Exercise 5. Let $p \in [0, 1]$ be a parameter. Consider the random walk induced by the 1-state pPDS over stack alphabet $\Gamma = \{X\}$ and transitions

$$X \xrightarrow{1-p} \varepsilon \quad \text{and} \quad X \xrightarrow{p} X^2.$$

Show that whether the probability of termination from configuration X is 1 depends on the value of p .

Exercise 6. Let $n \in \mathbb{N}$ and $p \in [0, 1]$ be parameters. Consider the random walk induced by the 1-state *pPDS* over stack alphabet $\Gamma = \{X\}$ and transitions

$$X \xrightarrow{1-p} \varepsilon \quad \text{and} \quad X \xrightarrow{p} X^n.$$

Show that the probability of termination x from configuration $X\varepsilon$ is 1 if, and only if, $p = 1/n$.

Probabilistic pushdown automata

Exercise 7. Show that the quantitative reachability problem for *pPDA* is decidable with *PSPACE* complexity. Hint: Use the fact that the existential fragment of Tarki's arithmetic is decidable in *PSPACE*.

Exercise 8. Can we decide efficiently whether $[pXq] = 0$? And whether $[pXq] = 1$?

References